Pavel Vasilyev

Senior AI Engineer - Agentic Systems

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PROFILE

Senior AI engineer with fifteen years across applied ML, NLP, and software. Currently building evaluation and routing infrastructure for LLM agents in Elixir and Python. Earlier work in NLP, quantitative research, and data science.

SKILLS

LLM systems: Agent evaluation, tool calling, provider routing and fallback, cost and latency tracing. Claude, Gemini, OpenRouter, Ollama, Cloudflare Workers AI.

Machine learning: NLP, text classification and extraction, feature engineering, model selection and training.

Languages: Elixir, Python, Rust, Swift, TypeScript, Clojure.

Infrastructure: Nix, Caddy, systemd, Git workflows.

EXPERIENCE

Independent Software Practice - consulting and open source 2010-present

Consulting, advisory, and software practice between salaried roles.

- Build LLM infrastructure: provider routing with fallback, retry and time budgets, and token and latency tracing. Build evaluation harnesses that score agent changes on pass rate, cost, and runtime.
- Design and deploy authenticated services and developer tooling in Elixir, Python, Rust, and Swift for clients and my own projects; 150+ projects across eight or more languages.

Research Lead - Spring Research 2020-2022

Quantitative research firm focused on financial-market modelling.

- Owned applied research on financial-market data end to end: new signal representations, feature engineering, model selection, and training.
- Communicated findings and collaborated as part of a working research team, carrying ideas from exploratory analysis into shared modelling work.

Earlier Data Science, NLP & Software Roles - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

Roles in Tel Aviv, Hong Kong, and Prague.

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer-vision pipelines; and corpus-linguistics infrastructure.

PROJECTS (PRIVATE REPOSITORIES, CODE ON REQUEST)

pi-harness - Evaluation harness for coding agents. Scores pass rate, cost, and runtime per run and A/B-tests configuration changes; one change raised spec-first compliance from 0/3 to 3/3 at +26% cost (n=3).

inference_harness - LLM job runner in Elixir: provider routing and fallback, retry and time budgets, token and latency tracing.

free-llm-harness - Python harness for testing LLM endpoints: backoff, model selection, tool calls. Measured success rate and p95 latency over 88 requests.

EDUCATION

Education: MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Languages: English, Hebrew, Russian; working knowledge of German and French.